Exercise Sheet 3 (theory part):

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Exercise 1: Principal Component Analysis (15 + 15 P)

We consider a dataset $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N \in \mathbb{R}^d$. Principal component analysis searches for a unit vector $\boldsymbol{u} \in \mathbb{R}^d$ such that projecting the data on that vector produces a distribution with maximum variance. Such vector can be found by solving the optimization problem:

$$\arg\max_{\boldsymbol{u}} \ \frac{1}{N}\sum_{i=1}^{N}\left(\boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m})\right)^2 \quad \text{with} \quad \|\boldsymbol{u}\|^2 = 1 \quad \text{and} \quad \boldsymbol{m} = \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{x}_i.$$

(a) Show that the problem above can be rewritten as

$$\arg\max_{\boldsymbol{u}} \ \boldsymbol{u}^\top \Sigma\, \boldsymbol{u} \quad \text{with} \quad \|\boldsymbol{u}\|^2 = 1, \qquad \text{where} \quad \Sigma = \frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{x}_i - \boldsymbol{m})(\boldsymbol{x}_i - \boldsymbol{m})^\top$$

is the covariance matrix.

Answer:

As $\boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m})$ is scalar we can write $(\boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m}))^2$ such that:

$$\left(\boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m})\right)^2 = \boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m})(\boldsymbol{x}_i - \boldsymbol{m})^\top\boldsymbol{u}.$$

Therefore,

$$\begin{aligned}\frac{1}{N}\sum_{i=1}^{N}\left(\boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m})\right)^2 &= \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m})(\boldsymbol{x}_i - \boldsymbol{m})^\top\boldsymbol{u} \\ &= \boldsymbol{u}^\top\left(\frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{x}_i - \boldsymbol{m})(\boldsymbol{x}_i - \boldsymbol{m})^\top\right)\boldsymbol{u} \\ &= \boldsymbol{u}^\top\Sigma\boldsymbol{u}.\end{aligned}$$

The constraint $\|\boldsymbol{u}\|^2 = 1$ is unchanged. Hence the original optimization problem is equivalent to

$$\arg\max_{\boldsymbol{u}} \ \boldsymbol{u}^\top\Sigma\boldsymbol{u} \quad \text{with} \quad \|\boldsymbol{u}\|^2 = 1.$$

(b) Show using the method of Lagrange multipliers that the problem above can be reformulated as solving the eigenvalue problem $\Sigma\boldsymbol{u} = \lambda\boldsymbol{u}$ and retaining the eigenvector $\boldsymbol{u}$ associated to the highest eigenvalue λ.

Answer:

We solve

$$\max_{\boldsymbol{u}} \ \boldsymbol{u}^\top \Sigma \boldsymbol{u} \quad \text{subject to} \quad \boldsymbol{u}^\top \boldsymbol{u} - 1 = 0.$$

$$\mathcal{L}(\boldsymbol{u}, \lambda) = \boldsymbol{u}^\top \Sigma \boldsymbol{u} - \lambda \left(\boldsymbol{u}^\top \boldsymbol{u} - 1 \right).$$

Taking the derivative with respect to $\boldsymbol{u}$ and setting it to zero gives

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{u}} = 2\Sigma \boldsymbol{u} - 2\lambda \boldsymbol{u} = 0.$$

Hence

$$\Sigma \boldsymbol{u} = \lambda \boldsymbol{u}.$$

So any stationary point of the optimization problem is an eigenvector of Σ. If we multiply both sides by $\boldsymbol{u}^\top$ and using $\boldsymbol{u}^\top \boldsymbol{u} = 1$, we obtain

$$\boldsymbol{u}^\top \Sigma \boldsymbol{u} = \lambda \boldsymbol{u}^\top \boldsymbol{u} = \lambda.$$

LHS of the equation ($\boldsymbol{u}^\top \Sigma \boldsymbol{u}$) is our objective which we want to maximize.

For the objective to be maximized, we should choose the eigenvector in a way that the corresponding eigenvalue λ is maximum. In other words, one should choose the leading eigenvector.

Exercise 2: Bounds on Eigenvalues (10 + 10 P)

Let λ_1 denote the largest eigenvalue of the matrix Σ. The eigenvalue λ_1 measures the variance of the data when projected on the first principal component. We study how the latter can be bounded with the diagonal elements of the matrix Σ.

Preliminary observation for all eigenvalues/eigenvectors $(\lambda_i, \boldsymbol{u}_i)$: show that $\lambda_i = \boldsymbol{u}_i^\top \Sigma \boldsymbol{u}_i \geq 0$.

Answer (preliminary observation):

Let $(\lambda_i, \boldsymbol{u}_i)$ be an eigenpair of Σ, so that

$$\Sigma \boldsymbol{u}_i = \lambda_i \boldsymbol{u}_i, \qquad \|\boldsymbol{u}_i\| = 1.$$

Then, take $\Sigma \boldsymbol{u}_i = \lambda_i \boldsymbol{u}_i, \qquad$multiply both sides from left with $\boldsymbol{u}_i^\top$:

$$\boldsymbol{u}_i^\top \Sigma \boldsymbol{u}_i = \boldsymbol{u}_i^\top (\lambda_i \boldsymbol{u}_i) = \lambda_i \boldsymbol{u}_i^\top \boldsymbol{u}_i = \lambda_i.$$

Using the definition of Σ,

$$
\begin{aligned}
\lambda_i &= \boldsymbol{u}_i^\top \Sigma \boldsymbol{u}_i \\
&= \boldsymbol{u}_i^\top \left(\frac{1}{N} \sum_{j=1}^{N} (\boldsymbol{x}_j - \boldsymbol{m})(\boldsymbol{x}_j - \boldsymbol{m})^\top \right) \boldsymbol{u}_i \\
&= \frac{1}{N} \sum_{j=1}^{N} \boldsymbol{u}_i^\top (\boldsymbol{x}_j - \boldsymbol{m})(\boldsymbol{x}_j - \boldsymbol{m})^\top \boldsymbol{u}_i \quad (\text{distribute } \boldsymbol{u}_i^\top \text{ and } \boldsymbol{u}_i \text{ into the sum}) \\
&= \frac{1}{N} \sum_{j=1}^{N} \left(\boldsymbol{u}_i^\top (\boldsymbol{x}_j - \boldsymbol{m}) \right) \left((\boldsymbol{x}_j - \boldsymbol{m})^\top \boldsymbol{u}_i \right) \quad (\text{group the terms}) \\
&= \frac{1}{N} \sum_{j=1}^{N} \left(\boldsymbol{u}_i^\top (\boldsymbol{x}_j - \boldsymbol{m}) \right)^2 \quad (\text{since scalar transpose } \boldsymbol{a}^\top \boldsymbol{b} = (\boldsymbol{a}^\top \boldsymbol{b})^\top = \boldsymbol{b}^\top \boldsymbol{a}) \\
&\geq 0.
\end{aligned}
$$

(a) Show that $\sum_{i=1}^{d} \Sigma_{ii}$ is an upper bound to the eigenvalue λ_1.

Hint: relate the trace of Σ to its eigenvalues.

Answer:

The trace of Σ is the sum of its diagonal entries:

$$\operatorname{tr}(\Sigma) = \sum_{i=1}^{d} \Sigma_{ii}.$$

Since Σ is symmetric, it has an orthonormal eigenbasis, and the trace equals the sum of its eigenvalues:

$$\operatorname{tr}(\Sigma) = \sum_{i=1}^{d} \lambda_i.$$

Therefore,

$$\sum_{i=1}^{d} \Sigma_{ii} = \sum_{i=1}^{d} \lambda_i.$$

From the preliminary observation, all eigenvalues are nonnegative, so

$$\sum_{i=1}^{d} \lambda_i = \lambda_1 + \sum_{i=2}^{d} \lambda_i \geq \lambda_1.$$

Hence

$$\sum_{i=1}^{d} \Sigma_{ii} = \operatorname{tr}(\Sigma) = \sum_{i=1}^{d} \lambda_i \geq \lambda_1.$$

So $\sum_{i=1}^{d} \Sigma_{ii}$ is an upper bound on λ_1.

(b) Show that $\max_{i=1}^{d} \Sigma_{ii}$ is a lower bound to the eigenvalue λ_1.

Hint: $\lambda_1 = \max_{\|\boldsymbol{u}\|=1} \boldsymbol{u}^\top \Sigma \boldsymbol{u}$; restrict the maximization to the canonical basis vectors $\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d$.

Answer:

By the variational characterization of the largest eigenvalue of a symmetric matrix,

$$\lambda_1 = \max_{\|\boldsymbol{u}\|=1} \boldsymbol{u}^\top \Sigma \boldsymbol{u}.$$

For every canonical basis vector $\boldsymbol{e}_i$, we have $\|\boldsymbol{e}_i\| = 1$, and

$$\boldsymbol{e}_i^\top \Sigma \boldsymbol{e}_i = \Sigma_{ii}.$$

Since λ_1 is the maximum over all unit vectors, it must be at least as large as the value obtained by any particular unit vector. In particular, for every i,

$$\lambda_1 \geq \boldsymbol{e}_i^\top \Sigma \boldsymbol{e}_i = \Sigma_{ii}.$$

Therefore,

$$\lambda_1 \geq \max_{i=1}^{d} \Sigma_{ii}.$$

So $\max_{i=1}^{d} \Sigma_{ii}$ is a lower bound on λ_1.

Exercise 3: Iterative PCA (5 + 10 + 5 P)

When performing principal component analysis, computing the full eigendecomposition of the covariance matrix Σ is typically slow, and we are often only interested in the first principal components. An efficient procedure to find the first principal component is *power iteration*. It starts with a random unit vector $\boldsymbol{w}^{(0)} \in \mathbb{R}^d$, and iteratively applies the parameter update

$$\boldsymbol{w}^{(t+1)} = \frac{\Sigma \boldsymbol{w}^{(t)}}{\|\Sigma \boldsymbol{w}^{(t)}\|}$$

until some convergence criterion is met. Here, we would like to show the exponential convergence of power iteration. For this, we look at the error terms

$$E_k(\boldsymbol{w}) = \frac{\boldsymbol{w}^\top \boldsymbol{u}_k}{\boldsymbol{w}^\top \boldsymbol{u}_1} \qquad \text{with} \quad k = 2, \ldots, d,$$

and demonstrate that they all converge to zero as $\boldsymbol{w}$ approaches the eigenvector $\boldsymbol{u}_1$ and becomes orthogonal to the other eigenvectors. To demonstrate this, we proceed in three steps:

(a) Show that

$$E_k(\boldsymbol{w}^{(t+1)}) = \frac{\boldsymbol{w}^{(t)\top} \Sigma \boldsymbol{u}_k}{\boldsymbol{w}^{(t)\top} \Sigma \boldsymbol{u}_1}.$$

Answer:

$$
\begin{aligned}
E_k(\boldsymbol{w}^{(t+1)}) &= \frac{\boldsymbol{w}^{(t+1)\top}\boldsymbol{u}_k}{\boldsymbol{w}^{(t+1)\top}\boldsymbol{u}_1} \\
&= \frac{\left(\frac{\Sigma\boldsymbol{w}^{(t)}}{\|\Sigma\boldsymbol{w}^{(t)}\|}\right)^\top \boldsymbol{u}_k}{\left(\frac{\Sigma\boldsymbol{w}^{(t)}}{\|\Sigma\boldsymbol{w}^{(t)}\|}\right)^\top \boldsymbol{u}_1} \\
&= \frac{\left(\Sigma\boldsymbol{w}^{(t)}\right)^\top \boldsymbol{u}_k}{\left(\Sigma\boldsymbol{w}^{(t)}\right)^\top \boldsymbol{u}_1} \\
&= \frac{\boldsymbol{w}^{(t)\top}\Sigma^\top \boldsymbol{u}_k}{\boldsymbol{w}^{(t)\top}\Sigma^\top \boldsymbol{u}_1} \\
&= \frac{\boldsymbol{w}^{(t)\top}\Sigma \boldsymbol{u}_k}{\boldsymbol{w}^{(t)\top}\Sigma \boldsymbol{u}_1}.
\end{aligned}
$$

The last equality follows because the covariance matrix is symmetric, i.e. $\Sigma^\top = \Sigma$. $\square$

(b) Starting from the result above, show that

$$
E_k(\boldsymbol{w}^{(t+1)}) = \left|\frac{\lambda_k}{\lambda_1}\right| \cdot E_k(\boldsymbol{w}^{(t)}).
$$

Hint: Recall that the covariance is linked to its eigenvectors and eigenvalues through the equation $\Sigma\boldsymbol{u}_i = \lambda_i\boldsymbol{u}_i$.

Answer:

Using the result from (a) and the eigenvalue equation, we obtain

$$
\begin{aligned}
E_k(\boldsymbol{w}^{(t+1)}) &= \frac{\boldsymbol{w}^{(t)\top}\Sigma \boldsymbol{u}_k}{\boldsymbol{w}^{(t)\top}\Sigma \boldsymbol{u}_1} \\
&= \frac{\boldsymbol{w}^{(t)\top}\lambda_k \boldsymbol{u}_k}{\boldsymbol{w}^{(t)\top}\lambda_1 \boldsymbol{u}_1} \\
&= \left|\frac{\lambda_k}{\lambda_1}\right| \frac{\boldsymbol{w}^{(t)\top}\boldsymbol{u}_k}{\boldsymbol{w}^{(t)\top}\boldsymbol{u}_1} \\
&= \left|\frac{\lambda_k}{\lambda_1}\right| E_k(\boldsymbol{w}^{(t)}).
\end{aligned}
$$

Since Σ is a covariance matrix, all its eigenvalues are nonnegative, and we assume $\lambda_1 > 0$. $\square$

(c) Starting from the result above, show that

$$
E_k(\boldsymbol{w}^{(T)}) = \left|\frac{\lambda_k}{\lambda_1}\right|^T \cdot E_k(\boldsymbol{w}^{(0)}),
$$

i.e. the convergence of the algorithm is exponential with the number of time steps T.

Answer:

We prove the statement by mathematical induction.

Base case ($T = 0$):

$$
\begin{aligned}
E_k(\boldsymbol{w}^{(0)}) &= \left|\frac{\lambda_k}{\lambda_1}\right|^0 E_k(\boldsymbol{w}^{(0)}) \\
&= E_k(\boldsymbol{w}^{(0)}).
\end{aligned}
$$

Thus, the statement holds for $T = 0$.

Induction hypothesis:

Assume that the statement holds for some $T \in \mathbb{N}_0$, i.e.

$$
E_k(\boldsymbol{w}^{(T)}) = \left|\frac{\lambda_k}{\lambda_1}\right|^T E_k(\boldsymbol{w}^{(0)}).
$$

Induction step ($T \to T+1$):

Using the result from (b) and the induction hypothesis, we obtain

$$
\begin{aligned}
E_k(\boldsymbol{w}^{(T+1)}) &= \left|\frac{\lambda_k}{\lambda_1}\right| E_k(\boldsymbol{w}^{(T)}) \\
&= \left|\frac{\lambda_k}{\lambda_1}\right| \left|\frac{\lambda_k}{\lambda_1}\right|^T E_k(\boldsymbol{w}^{(0)}) \\
&= \left|\frac{\lambda_k}{\lambda_1}\right|^{T+1} E_k(\boldsymbol{w}^{(0)}).
\end{aligned}
$$

Therefore, the statement also holds for $T + 1$.

By mathematical induction, we conclude that

$$
\boxed{E_k(\boldsymbol{w}^{(T)}) = \left|\frac{\lambda_k}{\lambda_1}\right|^T E_k(\boldsymbol{w}^{(0)})}
$$

for all $T \in \mathbb{N}_0$. $\square$